Cholesterol Learning Module

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Physician Learning Program

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Chapter 1: Updated Needs Assessment

Introduction

4th year Information Design students at Mount Royal University (MRU) have been presented with a design challenge for the Physician Learning Program (PLP), housed under the University of Calgary.

The design challenge is centered around how to improve the conversation between family doctors and patients who have high cholesterol. This includes how patients and providers can have a shared decision-making discussion around cholesterol management, where patients feel they are empowered and can confidently take control and ownership of their cholesterol health with the guidance of family providers.

Organizational Description

About

The Physician Learning Program (PLP) is driven to move from evidence to actionable outcomes. PLP does this by creating an environment where everyone is welcome, their voices are heard and collaborates to tackle the most pressing healthcare issues. By doing so, the PLP is well positioned to help create a world-class, responsive, and sustainable healthcare system. (AMA, n.d. para 1).

With access to extensive health data, along with diverse expertise in human-centered design, quality improvement and implementation strategies, PLP supports physicians and care teams by going from knowledge to action. (AMA, n.d., para 1).

Mission + Vision

The Physician Learning Program's mission is to create actionable clinical information and engage with physicians, teams, patients, and partners to co-create sustainable solutions to advance practice in the province of Alberta (Physician Learning Program, para. 1).

The vision that the Physician Learning Program holds is, by 2025, all Alberta physicians will care for patients in a supportive culture, driven by evidence-informed, reflective practice improvement (Physician Learning Program, para. 1).

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Values

The Physician Learning Program adheres to values that allow it to change and advance physician practice, based on the following principles:

- Human-centered design
- Implementation science
- Knowledge translation
- Supportive culture
- Reflective

Purpose

The aim of this project is to enable patients to have agency over their own, at-home care and treatment plan, focused on tackling the longstanding issue of high cholesterol. More often than not, patients with high cholesterol are averse to taking medication, and slip out of diet plans. Furthermore, most patients do not understand the impact that high cholesterol has on the human body and remain complacent until a health event occurs.

The project goal is to adequately inform patients of the risks of high cholesterol, its comorbidities, as well as arm them with valuable knowledge about diet planning and medication use.

Design Challenge

Problem Statement

The PLP organization has signaled that they experience significant hurdles in relation to informing and guiding patients to act on their health, and that they often do not take the diagnosis of high cholesterol seriously. A significant part of this challenge is that many physicians bring up elevated cholesterol levels as a side note, or gloss over the fact - understating its importance to one's health.

The primary goals within the scope of this project are:

- To inform the patient of the risks of high cholesterol
- To educate the patient about comorbidities and associated illnesses that are borne of high cholesterol
- Create accessible diet and lifestyle management plans
- Educate patients on the mechanism and safety of statin usage for treatment

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Considerations within the scope of the project are:

- Physicians have a limited amount of time to explain primary factors to patients
- Patients come from a wide variety of beliefs, lifestyles, and life factors
- Patients must be given easily digestible knowledge so that it is retained and followed and not overwhelming

Learning Objectives

The learning objectives of this community learning assignment project are two-fold.

- First, for students in the Information Design course to apply the skills they have been taught throughout their degree and use them towards a real world issue and using 21st century skill, such as collaboration, open communication in a world life scenario
- Secondly, it is for the Physician Learning Program team to use this needs assessment guide and use the proposed design solutions how they seem appropriate, regarding the issue of managing cholesterol and allowing patients to have the awareness to bring this "abstract" topic of cholesterol to their next appointment

Performance Gaps

PLP currently struggles with several factors in addressing high cholesterol from both patient and physician standpoints. The chief performance gaps for physicians are:

- Lack of time to elaborate on issues and connect with the patient
- Difficulty in knowledge translation and communication
- Difficulty communicating the risk factors in a human-centered design approach

The PLP has also noted that patients present with performance gaps, as follows:

- Patients do not understand the risks of high cholesterol
- Patients are averse to medication, despite the relative safety and low cost of statins
- Patients can struggle with long-term, impactful lifestyle changes for a multitude of factors, including but not limited to: time, work schedule, present life complications, health limitations, cost, attitudes towards health, etc.



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Project Constraints

The PLP has communicated that the primary goal of the project is accessibility and easy implementation. This means that a solution must be highly compressed as well as easily distributable, with minimal investment cost to counter it. Some notable factors of this constraint include:

- Any website or application must not have a log-in, due to security and bureaucratic access being scarce to acquire.
- Any print solutions must be attention grabbing, and easily distributable and printable by physicians.
    - » Printed materials must meet accessibility guidelines, meaning that they must have easily read text and/or a dyslexia-friendly font, grey scale legibility, amply sized lettering and kerning, proper colour contrast, as well as easily interpretable iconography.
- The up-front and investment cost for implementation must be kept to a minimum.

Access Points

Access points are areas of entry that lead to solutions. Through identifying potential demographics that struggle with high cholesterol as well as their associations and potential views, several access points have been identified. These are potential take-off points that materials may centre around, allowing the project to build on demographics or groups that may act as a base group for assessment. The following access points have been identified:

- 40+ age group
- Trades workers
- Busy physicians in need of easily understandable materials

Desired Outcomes

The probable and desired outcome of this project is to successfully educate patients and budge them into actionable choices over their health, as well as give physicians a human-focused approach and materials to allow them to translate their knowledge into terms and outcomes that a patient can easily understand. Further goals of this project include the successful production of a broad-spanning, human-centered, and digestible information piece prototype that would positively impact the health of patients in Canada.

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Personas

Target Audience

The target group that we are focusing on in this project are two diverse groups, both affected by this problem of addressing cholesterol.

The first group are Albertans that are 40+ years of age with varying levels of health literacy, who are at high risk or are already living with high cholesterol. These are individuals whose lab work shows that they are currently at risk or already have high cholesterol, due to many factors, and would benefit greatly from taking a statin to manage their cholesterol.

The second group is the physician in a community family medicine practice. These doctors are often seen as the ones that see patients throughout a patient's lifetime and communicate and help patients manage and maintain their health before requiring specialty care. These physicians follow their patients through their journey and are there to help and provide meaningful and impactful information to their patients.



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Persona 1



https://fittopass.espyr.com/truck-driver-pros-and-cons/

Name

Tim Holesterine

Age

47

Occupation

Long distance truck driver

Place of Residence

Red Deer, AB

What Tim wants you to know about him:

- Tim always wanted to be a police officer, but unfortunately due to family circumstances, he became a truck driver instead.
- While he understands that taking good care of his health is a good choice, he has a hard time understanding the complexity and the impact cholesterol can have on his life. When he is on the road, he finds it a challenge to find healthy options.
- When he talks to his family doctor, the conversation becomes one-sided and the doctor talks too technically and Tim doesn't understand what he is telling him.
- He knows that medication might be useful, he is afraid as his family has had bad experiences with medicine and doesn't want to fall into what he sees as a trap.

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Persona 2



https://www.comservice.bc.ca/our-program-categories/seniors/

Name

Sally Goods

Age

60

Occupation

Retired

Place of Residence

Calgary

What Sally wants you to know about her:

- Sally was a former elementary school teacher and knows the importance of knowledge and education.
- She has pristine health and has no major health challenges.
- She has been informed that she has above average LDL levels, but is willing to consider taking medication, but doesn't really know all that much about what exactly cholesterol does.
- Wishes that her family doctor had an appointment dedicated to helping her understand what cholesterol means for her, personally.

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Persona 3



https://www.menshealth.com/health/a40971698/5-questions-your-doctor-wishes-youd-ask/

Name

Dr. Peter Ross

Age

31

Occupation

Family Physician

Place of Residency

Edmonton, AB

What Peter wants you to know about him:

- Deeply cares about his practice and the patients that put their trust in him.
- He has a busy schedule daily, allotting, on average, 15 minutes per patient.
- Has tried many strategies to ensure patient concerns are met and cover everything that is deemed important, but unfortunately there isn't enough time as he is billed by a fee for service model.
- Wish patients would understand how deadly LDL cholesterol can be if left untreated.

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Considerations

Time Constraints

The PLP association has identified that their deadline for the scope of this project ends in early December, allowing us sufficient time to create a needs analysis as well as develop a rough prototype based on current analytical and empathetic understanding of the issue at hand.

Skills

There are a multitude of skills needed to accurately and thoughtfully craft a solution within the scope of this project. The primary and vital factors will allow our team to meet our goal as well as design a prototype mindful and adherent to regulations, project constraints, and positively impact patient health. The factors necessary to meet our goals are as follows:

- Empathetic understanding
- Analytical approach to related systemic practice
- Instructional design
- Human-centered design
- Collaborative approach
- Understanding of critical health factors

Behaviours

Many different attitudes towards health will be represented within the given demographic constraint of 40+, meaning that any solutions must be broadly accessible, understandable, and legible. This includes immigrants, busy workers, those with poor health knowledge, as well as those who rarely seek physician access.

These factors must all be taken into consideration, as they have a significant impact on how a patient will perceive, assess, understand, translate, and implement information input and any practices given to them.

Internal and External Human Factors

Internal (physicians) and external (patients) factors are a vital aspect of success within this project. Internal resources, training, and empathetic understanding must be channeled, but cannot be assumed. Not all physicians will have the ability to translate knowledge or connect with patients, meaning that any prototype created must contain empathetic but concrete descriptions and terminology translated to public terms.

External factors related to patients must be considered in terms of appropriate and adequate

information, terms, as well as the accessibility of the document. The patient must leave the information piece with gained information, and potentially a resource to consult should they have additional questions. Patients must also be able to understand the prototype in a meaningful way, meaning that it must have an impact on their attitude towards their health.

Current Resources

Patient Resources

These are the current patient resources used when informing patients of and about high cholesterol and the risk of it.

myHealthAlberta - High Cholesterol

https://myhealth.alberta.ca/health/pages/conditions.aspx?Hwid=hw115432

Healthline - High Cholesterol (US Based)

https://www.healthline.com/health/high-cholesterol

HeartandStroke.ca – Managing Cholesterol

https://www.heartandstroke.ca/heart-disease/risk-and-prevention/condition-risk-factors/managing-cholesterol

Cardiovascular Risk Calculators

A variety of risk calculators are often used, the first one is analogue, which will help to understand how the scores are determined for the visual calculators. The Framingham risk scores are validated scores of risk over a period of time.

Academic Papers

The academic papers focus on why physicians and patients struggle with getting people to start using a statin, and what helps to get people on a statin. As well as, they also highlight the use of shared decision-making.

Clinical Guidelines

There are guideline documents used by primary care physicians to find what the appropriate treatments are for high cholesterol. The documents are the building blocks for the clinical decision-making processes and appropriate drug therapies for patients.

Approach

Learning Strategies + Methods

The design of this course is based on Distributed Learning theory, where the course is built for learning-on-demand and learner-centred instruction and training. This learning model allows for learners to learn remotely and independently where they choose the time and place to complete the course. This model focuses primarily on empowering learners to learn effectively and at their own pace, and the instructor (physician) is there to facilitate and help them along their way. Distance Learning theory was chosen because it aims to create open, flexible and distributed learning environments for learners to develop their knowledge and understanding of topics. By having a unique challenge at hand, distance learning allows for the patients to take home a course and learn at their own pace in a space they feel comfortable, as opposed to having them try and learn the information in the physician's office.

By providing small chunks of important information to the patients through digital means, with the option to print, the patient is given the opportunity to self direct their learning by being provided with tools to understand the risks and importance of taking care of their cholesterol at home. The recommended interventions will go hand in hand with in-person patient appointments, where the physician can inform and discuss health issues related to high cholesterol, continued risk assessment and possible plans of action. The course will be given to the patients to do at home, and it will provide easily digestible information about cholesterol and recommended treatment options for moving forward. The course will provide information that informs the patients on various aspects of cholesterol and cholesterol health.

There will be no data collected so that patients' privacy is protected, and they have the option to print out the information given to them at the end of the course so that they can continue their treatments when they are no longer in the clinic with their physicians.

Recommended Interventions

Module based Course

The module based course would be provided to people that are identified by their family physician, through discussions and by a standard and routine lab test for high cholesterol and/or those at risk of developing high levels of low-density lipoprotein (LDL). The modules would guide the participants through a series of related themes about cholesterol by educating them through small sections of information such as text, visuals and videos, to get people to understand the concept at hand. At the end of each module, participants have the opportunity to check their knowledge to ensure they understood the topic. Upon completing the modules, there is a mandatory quiz at the end, in which participants would need to score at least 80% on to finish the course and to ensure they understood the content given. If they get below an 80, they can repeat the quiz as many times as needed, until they get 80% or above. Furthermore, there are resources to help them navigate the healthcare system

regarding access to statin medication, if they are covered or uninsured, as well as logbooks - allowing them to document and track their diet and exercise for a certain period of time.

Information Printable: Logbooks

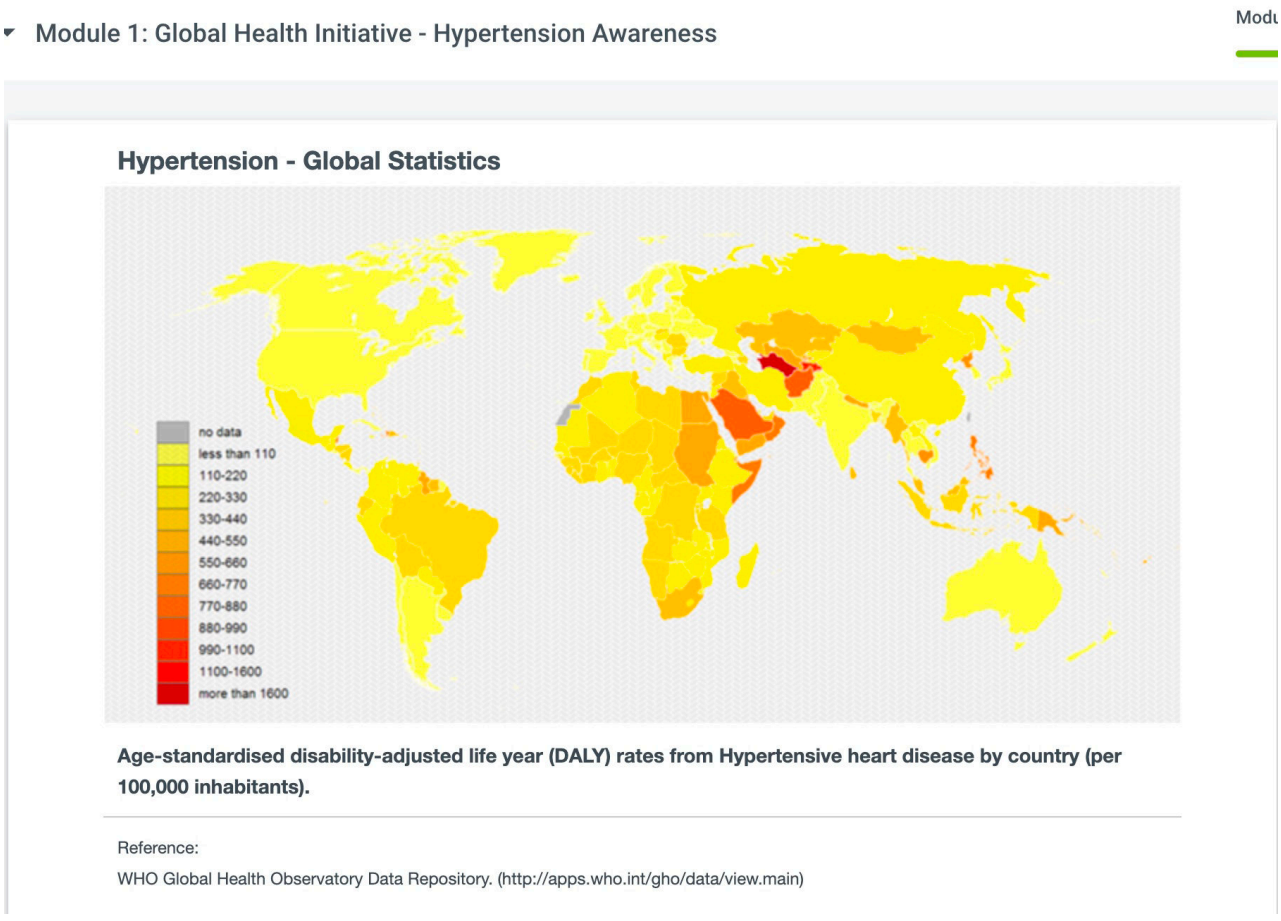
This is attached to the end of the course and is printable. It includes ways to track their diet and exercise routines for a week, so when they go to their doctor for a follow–up, they can have a shared decision-making appointment, where they discuss options and potentially the doctor may recommend statin intervention based on their lifestyle choices.

General Information About High Cholesterol

This would be included on the homepage of the self-assessment and can be printed off by users. It would be available to users to refer to after their doctor's visit and can be provided to them by family physicians.

Design Precedents

Global Health Initiative - Hypertension Awareness



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Diabetes – the Essential Facts (Coursera)

What is diabetes?
Graded Quiz • 30 min • 3 total points

Due Dec 18, 11:59 PM PST

1. Untreated diabetes is associated with … **1 point**

- ◯ Excessive physical activity
- ◯ High insulin production
- ◯ High glucose level
- ◯ Low heart rate

2. What is essential for regulating glucose levels in body and is produced in the Pancreas? **1 point**

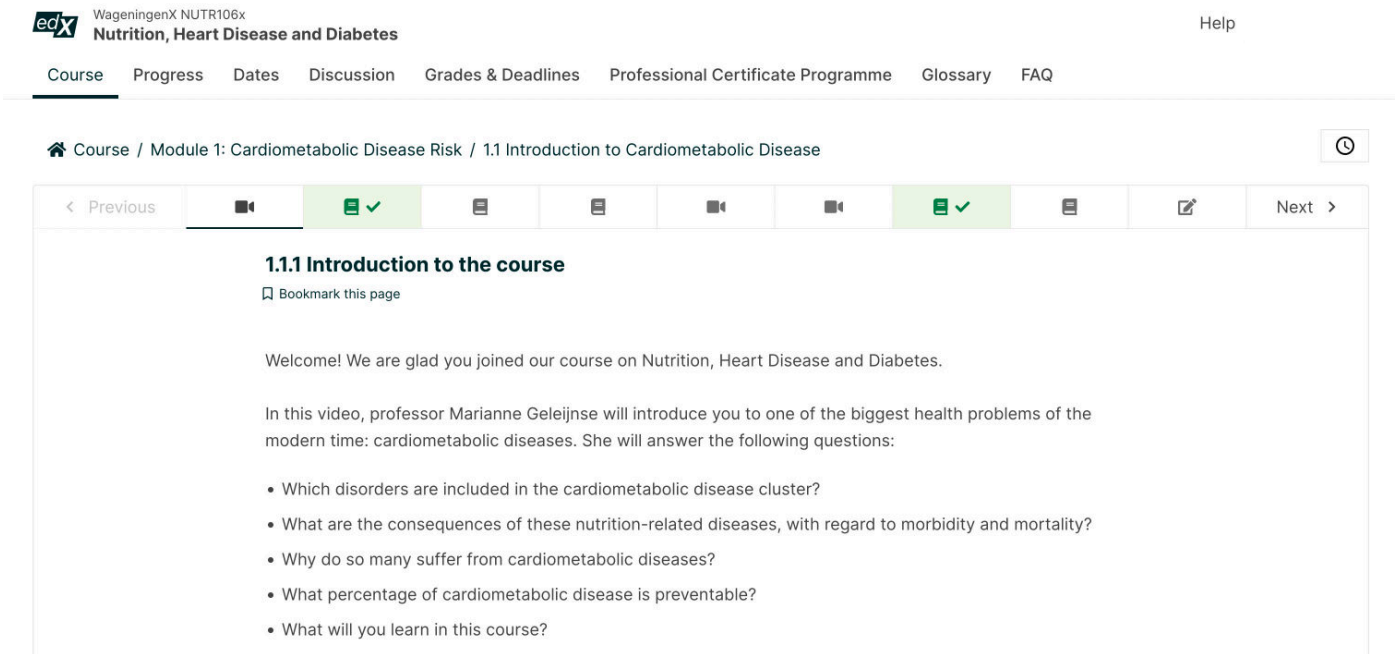
- ◯ Growth hormone
- ◯ Diabetes
- ◯ Glucose
- ◯ Insulin

3. Maintaining sufficient blood glucose levels is important. Which organ in the human body has the ability to take up glucose and produce glucose? **1 point**

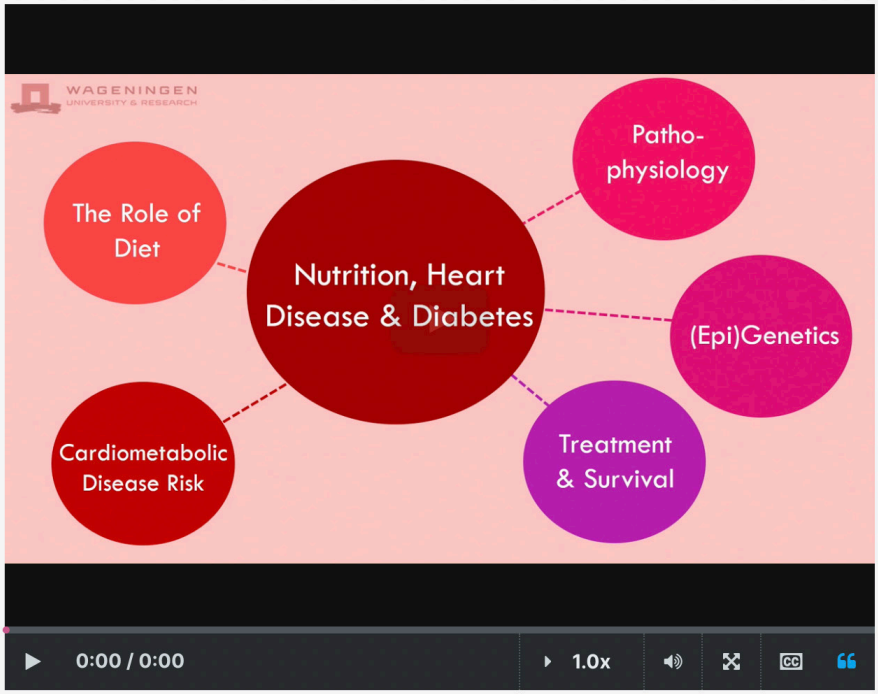
- ◯ The liver
- ◯ The stomach

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Nutrition, Heart Disease and Diabetes (edX)



Video: Introduction to the Course



Let me introduce you to one of the biggest public health problems

of the modern time,

called cardiometabolic diseases.

I am talking about a group of nutrition-related disorders

that form the largest contributor to mortality worldwide,

both in men and women.

They kill almost 18 million people each year,

which is around one-third of all deaths.

Not only in western societies, like the United States, Australia and Europe,

but also in Asia and emerging economies in Africa.

Cardiometabolic diseases are not just related to excess mortality.

They are also a major cause of morbidity,

which comes with hospitalisations, medical therapies

Video

Download video file

Transcripts

Download SubRip (.srt) file

Download Text (.txt) file

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Chapter 2: Executive Summary

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Executive Summary

For this project, we developed a module based learning experience. This learning module consists of educational elements by translating medical knowledge and terminology into ways that are easy to understand and comprehend, through a human-centered design lens. The course includes images, videos, short easy to read portions and test your knowledge sections. These elements come together to create a course to help those seeking applicable health knowledge to a current diagnosis, to have a safe space to learn. The cholesterol learning module aims to educate patients in an understandable, user focused way, to develop their knowledge about their cholesterol health, and encourage patients to be action oriented towards their cholesterol management in a way that is empowering. Our module is unique, as many currently available sources do not go as far as to define terms and ideas, and instead just present general information. The module focuses on the main elements of Distributed Learning Theory, where people are asked to take a self-guided course. Learners finish the modules by completing a quiz, based on the presented material they learned, in order to assess their retention of information, as well as their understanding of basic cholesterol and health strategies.

Concept

Problem:

Family doctors do not have adequate time to talk to their patients about the risks of high cholesterol

Doctor tells patient they have high cholesterol ▶ **Doctor "prescribes" them to complete eLearning course** ▶ **Patient completes course** ▶ **Patient follows up with doctor**

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Learning Module

The learning module contains topic descriptions and cursory glances at related issues, including comorbidities, and what cholesterol is, providing an overview of general facts before further educating the learner on best-practices for their cholesterol management. Furthermore, the module contains sections based on food, lifestyle, and medication, giving a summary on the most crucial lifestyle aspects as well as disproving commonplace myths about cholesterol and statins. Medication, exercise and diet, respectively, have the most significant impact when combined, and as such, represents a large portion of the educational material. Medication is covered in a specifically crafted section, designed to mitigate the risk of a patients' potential perception of what they may view as negative or frightening; and instead, educates them by drawing comparisons from applicable real world substances, and from those that have been in similar situations before starting statin medication. Furthermore, it also highlights the importance of medication potencies, as well as the safety of statins, specifically.

With regard to patients and human-centered design, the module is crafted to pinpoint patient concerns in a layman, easy to understand manner, wherein the patient may acquire a realistic understanding, and leave with a better developed understanding of what cholesterol is and ways to manage their cholesterol, as well as an incentive to follow-up with their primary care provider. This allows patients the ability to be an active participant in their care, creating a shared decision-making environment by having discussions with their doctor about their possible options to manage their cholesterol. This allows patients to continue to learn within their follow-up discussions, gain new confidence about their cholesterol, as well as gaining newfound health literacy along the way.



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Chapter 3: Course Example

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Lesson 1 of 9

Overview

Overview

In this course, you will be addressing the following:

1. Understand what cholesterol is and how that plays with your overall health
2. Learn how various treatment options can influence your cholesterol management
3. Learn, understand and identify how comorbidities play a part in cholesterol management

CONTINUE

On a scale of 1-5, how much do you know about cholesterol? (Select One)

- [ ] 1 (Very Little Knowledge)
- [ ] 2
- [ ] 3
- [ ] 4
- [ ] 5 (Very Knowledgeable)

SUBMIT

CONTINUE



Physician Learning Program

Lesson 2 of 9

What is Cholesterol?

Watch this video



Test your knowledge

Transports "bad cholesterol" back to the liver, which disposes it properly.

High-Density Lipoprotein (HDL)

Low-Density Lipoprotein (LDL)

CONTINUE

Physician Learning Program

The Effect of Diet



https://www.helpguide.org/articles/healthy-eating/healthy-eating.htm

Why Diet is Important?

Eating a nutritious, balanced diet is imperative for overall good health. More than exercise, what you eat is what you are.

DID YOU KNOW?

Fat and cholesterol can't dissolve in water or blood. Instead, the body packages fat and cholesterol into tiny, protein-covered particles called lipoproteins. These lipoproteins can transport a lot of fat, as they mix easily with blood and flow with it. Some of these particles are big and fluffy, while others are small and dense. This is what makes cholesterol. However, people who eat a diet of high sodium, high fat meals often contribute to LDL - which can lead to an increase risk of stroke, cardiovascular events amongst other events.

Does a hamburger contribute to high levels of LDL?

Yes

No

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Lesson 4 of 9

Exercising



https://www.earth.com/news/its-not-how-much-but-how-often-we-exercise-that-counts/

Why Exercise?

Exercise is good for you in general. But if you have issues with your cholesterol, it is one of the single most important things you can do to change your cholesterol health.

This is because as you move, your body burns energy and lowers your cholesterol.

Test your knowledge

Does exercise help with managing "bad" cholesterol?

Select yes or no.

Yes

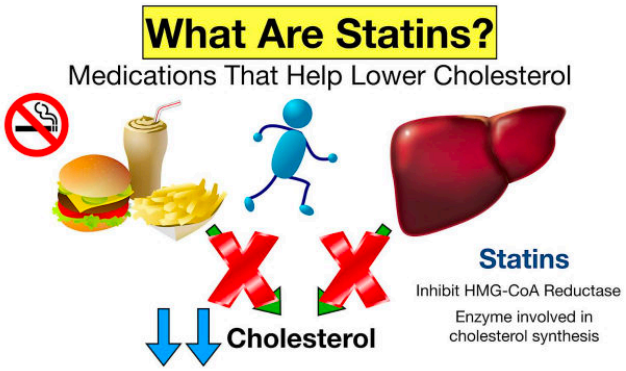
No

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Medications

About Statin Medications

Statin medications work by inhibiting (stopping) the mechanism which produces cholesterol in the body. This medication is known to be effective and safe, with major side effects being rare. It is compatible with a wide range of other medications. Additionally, statins are mostly covered by the Alberta government, reducing your out-of-pocket expenses.



Statins stop cholesterol from being synthesized. This diagram illustrates its scientific mechanism. To make it simple: a statin stops *any* cholesterol from being formed by your body by inhibiting the chemical mechanisms which form it.

Why take a statin medication?

Statin medications are a vital part of treating high cholesterol. Combined with lifestyle changes, it can safely and effectively reduce LDL cholesterol levels within patients. Some typical side effects of taking statins include:

Potential Side Effects

Statins, like any other medication, can affect everybody differently. Remember, you may not get any side effects at all, or just a few. Discuss this possibility with your doctor when considering any medication.

- Headache
- Dizziness
- Feeling unusually tired or physically weak
- Digestive system issues: constipation, flatulence, stomach upset
- Muscle pain
- Sleep problems

Of course, all of these side effects make statins sound scary.

Keep in mind, however, common OTC cough syrup has a similar list of side effects - lifesaving medications may be daunting, but they are not as scary as they can seem.

Statins are a safe method of reducing your cholesterol, and are fully covered by Alberta Health Services - they are free to Alberta residents. Additionally, statins are placed into severity categories - low intensity and high intensity have different potential for side effects, but come with the trade-off of lowering your cholesterol levels slower or faster, respectively.



"I'm worried about being on a medication for the rest of my life."

- Common Response to Statins

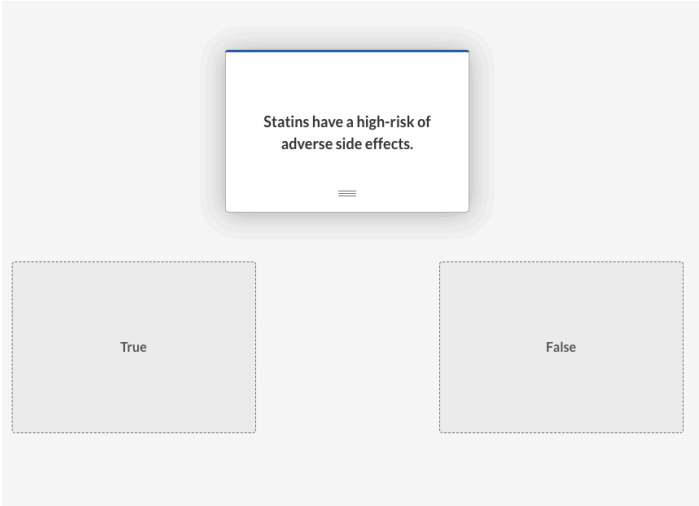
This is a reasonable thought that many people considering medications experience. Of course, it is not easy to face health issues. However, it is important to remember that statins are a lifesaving medication which can prevent critical health issues like strokes and heart attacks, and significantly lower your risk of another if you have already experienced a health event. For the vast majority of patients, the benefits outweigh the risks.

Statins can be taken on a flexible schedule, on your practitioner's recommended dose and time interval. You can ask for a weekly regimen, or alternate day regimen. Additionally, statins work best when taken overnight on an empty stomach - meaning that you can simply take one before bed without worry.

Speak to your doctor about testing the waters with a low-dose or low-potency statin. These low-potency statins have a lower chance of negative side effects - consistently taking a low-dose or low-potency statin can have great impact on your cholesterol, especially when combined with a statin-rich diet and exercise habits.

Test Your Knowledge

Drag the card into True or False categories based off of what you think is the right answer.

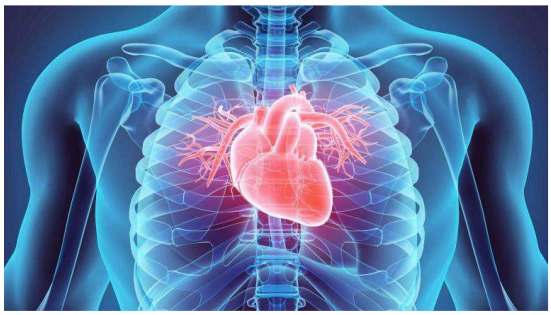


CONTINUE

Physician Learning Program

Lesson 6 of 9

Comorbidities



https://www.nhlbi.nih.gov/news/2011/conquering-cardiovascular-disease

Comorbidities

A comorbidity is when you have more than one condition at the same time. Unfortunately, comorbidities are usually chronic, meaning that they need long-term treatment or management.

While possible, high cholesterol is often connected to a commonality. This means it can appear with other chronic conditions.

Having comorbidities increases your risk of complications, including having high levels of LDL

Source: https://www.healthline.com/health/high-cholesterol/is-high-cholesterol-a-comorbidity#definition

Test Your Knowledge

Drag the card into the category based off of what you think is the right answer.

Yes

Does comorbidities influence cholesterol

Does managing comorbidities influence bad cholesterol risk

CONTINUE

Physician Learning Program

Lesson 7 of 9

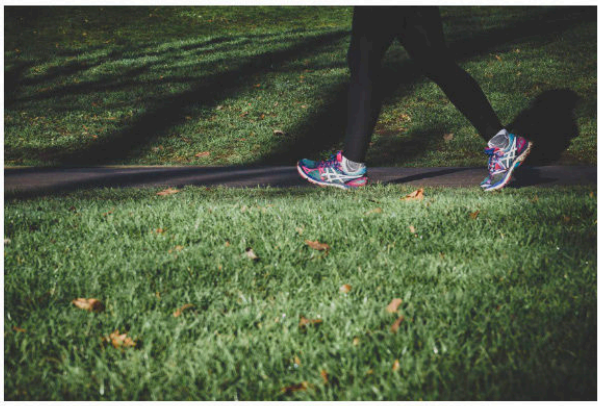
Test Your Knowledge

START QUIZ >

Question

01/05

True or False: Exercising regularly can help lower your risk of high cholesterol



- ○ True
- ○ False

SUBMIT

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Question

02/05

Are there two types of cholesterol?



- ○ Yes
- ○ No

SUBMIT

Question

03/05

What are some comorbidities that can lead to high levels of low-density lipoprotein. PICK THE BEST OPTION



- ☐ Type 2 Diabetes
- ☐ Kidney Disease
- ☐ High Blood Pressure
- ☐ All of the above

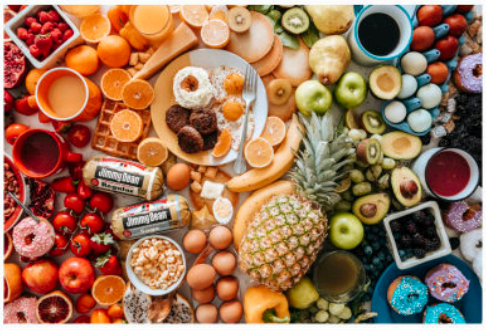
SUBMIT

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Question

04/05

Does diet help to contribute to managing your cholesterol levels? In particular, your LDL levels?



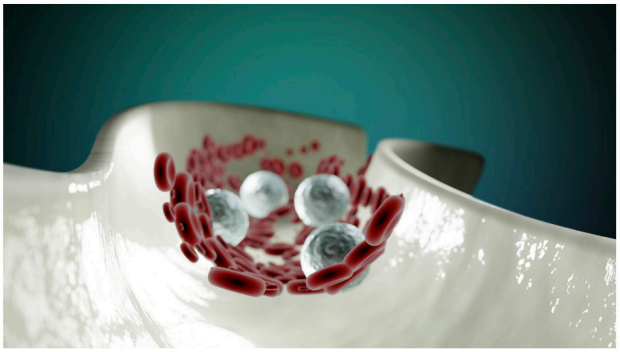
- ○ Yes
- ○ No
- ○ Maybe

SUBMIT

Question

05/05

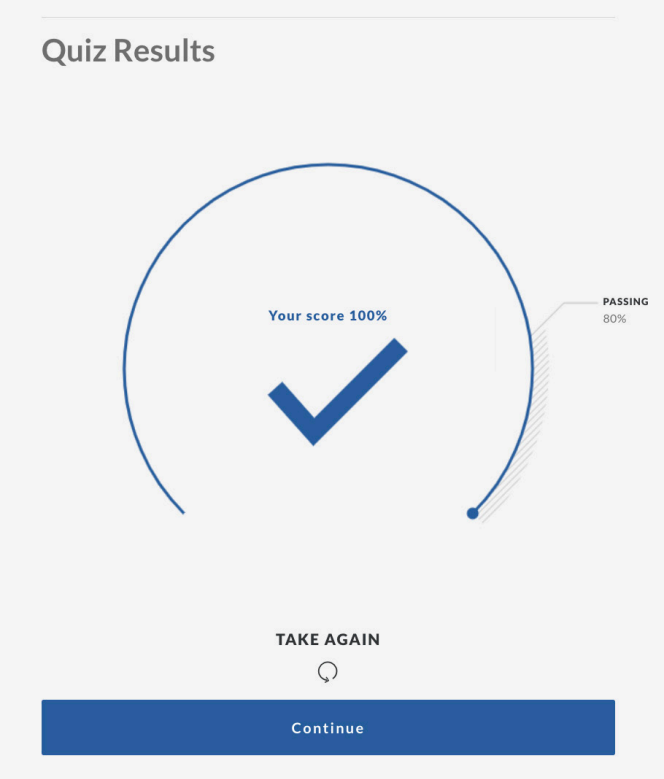
What is considered good cholesterol and bad cholesterol?



Low-Density Lipoprotein	Good Cholesterol
High-Density Lipoprotein	Bad Cholesterol

SUBMIT

Physician Learning Program



Lesson 8 of 9

Things You Can Do Next

Track Your Diet

Over the next week, you can track the things you eat and bring it to your next appointment to help discuss things you are doing well, and ways to improve.

Use the below took to help you keep track of your diet over the next week.

PDF Meal Log.pdf
63.1 KB

CONTINUE

Physician Learning Program

Weekly Meal Log

Date: _____

Day: Su M Tu W Th F Sa

Meal	What Did You Eat?	Notes
Breakfast		
Lunch		
Dinner		
Snacks		

Date: _____

Day: Su M Tu W Th F Sa

Meal	What Did You Eat?	Notes
Breakfast		
Lunch		
Dinner		
Snacks		

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Exercise Log

Over the next week, you can track your daily exercise and bring it to your next appointment to help discuss things you are doing well, and ways to improve.

Use the below took to help you keep track of your exercise over the next week.

PDF

Exercise Log.pdf

66 KB

CONTINUE

Physician Learning Program

Weekly Exercise Log

Date: _____

Day: Su M Tu W Th F Sa

Exercise Type	Time (mins)	Notes

Date: _____

Day: Su M Tu W Th F Sa

Exercise Type	Time (mins)	Notes

Date: _____

Day: Su M Tu W Th F Sa

Exercise Type	Time (mins)	Notes

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Look Up Statin Medication Coverage

In the chart below, you can find commonly prescribed statins that help lower your cholesterol. The chart includes the drug identification number (DIN) to help you check your health insurance to see how much is covered.

Drug Name (Brand)	Drug Identification Number (DIN)	Drug Name (Generic)
Lipitor	02230711 (10 mg) 02230713 (20 mg) 02230714 (40 mg) 02243097 (80 mg)	Atorvastatin
Lescol XL	02250527	Fluvastatin
Mevacor	00795852 (40 mg) 00795860 (20 mg)	Lovastatin
Pravachol	00893749 (10 mg) 00893757 (20 mg) 02222051 (40 mg) 02250330 (80 mg)	Pravastatin
Crestor	02265540 (5 mg) 02247162 (10 mg) 02247163 (20 mg) 02247164 (40 mg)	Rosuvastatin
Zocor	00884332 (10 mg) 00884340 (20 mg) 00884359 (40 mg)	Simvastatin

Next, you can see your health insurance provider's drug lookup tool to check your coverage.

Health Insurance Company	Drug Coverage Lookup
Alberta Blue Cross	Interactive Drug Benefit List (iDBL)
Sun Life	Check Your Coverage
Green Shield Canada	Online Services
Canada Life	Customer Sign In
Wawanesa Life	Customer Sign In
Desjardins	Group Insurance Individual Insurance
Group Medical Services (GMS)	Customer Login

CONTINUE

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I don't have coverage

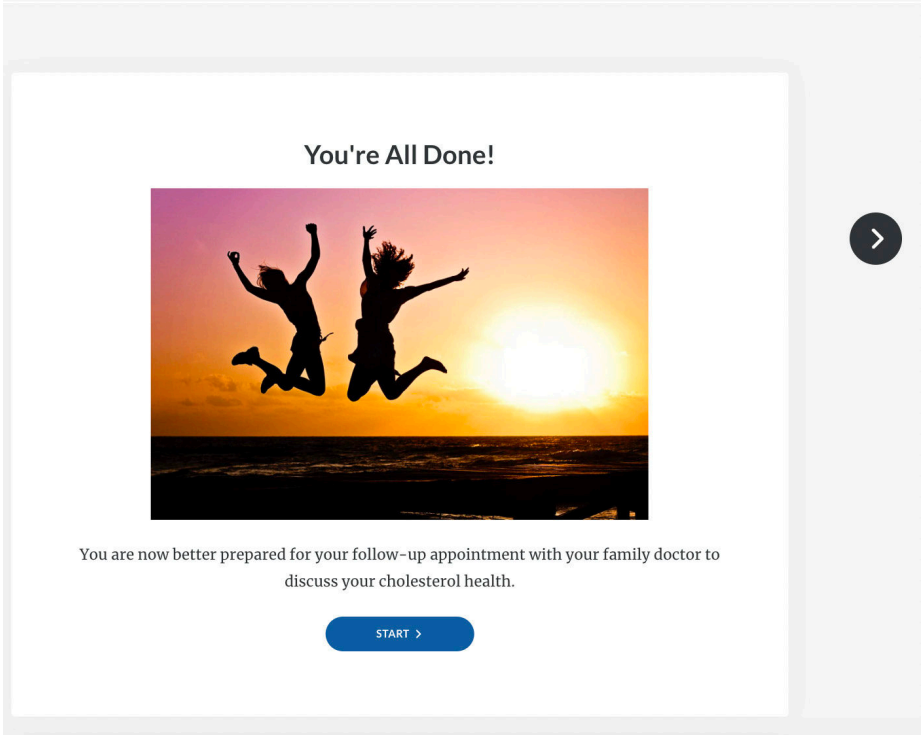
There are some free services available that may be able to assist you with a portion of your prescription costs. There are many options not shown, so be sure to look up if your prescription has a patient assistance plan.

- InnoviCares
- RxHelp.ca
- Zypitamag Savings Program
- Crestor Savings Program

CONTINUE

Lesson 9 of 9

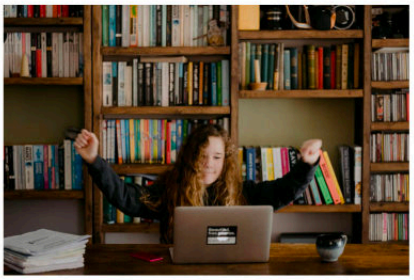
Wrap Up



Physician Learning Program

Step 1

Things You Have Learned



- What cholesterol is and the different types
- How exercise plays a role in managing your LDL cholesterol
- Why diet is an important management option
- How comorbidities play a factor in your cholesterol health
- The positive impacts of being on statin medications

1 ✓

CONTINUE

Next Steps



Call your family doctor to book your follow-up appointment to discuss a treatment plan for you and next steps for your cholesterol health.

START AGAIN

1 ✓

CONTINUE

Physician Learning Program

Check In

Answer the following questions to help us understand how useful this course was for your learning.

How confident are you now after taking this course with your knowledge on cholesterol? (Select One)

- [ ] 1 (Not Confident)
- [ ] 2
- [ ] 3
- [ ] 4
- [ ] 5 (Very Confident)

SUBMIT

CONTINUE

👋 Bye!

You may now leave this page.

Physician Learning Program

Chapter 4: Final Report

Desired Outcomes

The probable and desired outcome of this project is to successfully educate patients, and gently nudge them into actionable choices regarding their cholesterol health. Another desired outcome is to give physicians a human-focused approach, and materials, to allow them to translate their medical knowledge into terms and outcomes that a patient can easily understand and have an engaging and informative appointment. Furthermore, the goals of this project include the successful production of a broad-spanning, human-centered, and digestible information piece prototype, that would positively impact the health of patients in Canada. In future iterations, we would include this module based learning in different languages, so that individuals, whose native language is not English, can participate and take greater control of their cholesterol and actively be engaged in their overall health. Additionally, we would include a universal design lens - including options to change font sizes, and possibly text colour, so that patients who may have difficulties reading text online would be able to absorb the information the same as others. Furthermore, we would look into potentially adding an option to have the modules read aloud to the patients so that they can learn the information in an audiobook sort of style, for those who may not have the time to sit down and do the courses during their busy day-to-day lives. These additions would allow for a vast number of patients to be able to learn the importance of their cholesterol health and the options that are available to them.

Values

The Physician Learning Program adheres to values that allow it to change and advance physician practice, based on the following principles:

- Human-centered design
- Implementation science
- Knowledge translation
- Supportive culture
- Reflective

Physician Learning Program

Target Audience

The target audience that we are focusing on for this project are two distinct groups, both affected by this problem of addressing cholesterol health and understanding its importance. The first segment identified are Albertans that are 40+ years of age, with varying levels of health literacy, who, based on their lab work, are at high risk/ are already living with high cholesterol levels due to numerous factors. This segment identified would benefit greatly from lifestyle changes, as well as taking a statin to manage their cholesterol. The second segment identified are family physicians in a community family medicine practice. These doctors are often seen as the ones that see patients from cradle to grave, and are the touchpoint for patients; communicating and helping patients manage and maintain their health before requiring specialty care, such as cardiology or endocrinology care. These physicians follow their patients throughout a patient's health journey and are there to help, and provide meaningful and impactful information, as well as holistic care to their patients along the way.

Primary care providers distributing module access or information have a new way of delivering important information that may take up to several appointments to convey, to ensure that the patient and the provider have an understanding of expectations and action items to achieve in a patient friendly understandable way. This would as well cut down the amount of time physicians must spend repeating information to patients to ensure they understand the importance of cholesterol health. Primary care providers will be given the ability to have a more informed patient and have a targeted visit, as well as a way to incentivize a follow-up appointment for a plan of action for high cholesterol, helping to foster a healthier and trusting patient-doctor relationship. By having patients complete this course, the patients and their physicians will be able to have a more informative and comprehensive discussions with one another about cholesterol and its management.

Behaviours

Many different attitudes towards health will be represented within the given demographic constraint of 40+, meaning that any solutions must be broadly accessible, understandable, and legible. This includes immigrants, busy workers, those with poor health knowledge, as well as those who rarely seek physician access. These factors must all be taken into consideration, as they have a significant impact on how a patient will perceive, assess, understand, translate, and implement information input and any practices given to them.

Implementation

The implementation process is the steps in which this course will be provided to physicians as well as their patients. This process will allow for a seamless implementation of this course into physician's offices and get the course into patients hands.

Implementation of this cholesterol information course involves a four-pronged approach. These include:

- Getting the Course Ready for Distribution
- Training the Instructor
- Preparing the Learner
- Arranging a Learning Space - in the convenience of a patient's home or place of work

Getting the Course Ready for Distribution

The example course that was created was made using Articulate 360. This tool is easy to customize for the needs of this project and is flexible for adjustments that may need to be made. It is also easy to use and navigate for someone who has not designed a course before.

The course should also go through a trial period where it is given out and tested for 3 months, in which during this period any changes that need to be made can be done so that the course is in its best possible condition to be released.

Customizing the Course

The course we have created can be customized to better fit the needs of the Physician Learning Program at any point in time, allowing for adjustments to be made as problems or obstacles arise. This allows the course to be flexible and easily updated to provide the best quality up-to-date information and ensures the highest level of success for the Physician Learning Program.

Sharing the Course

The course can be shared using a link, which could be turned into a QR code that can be given to physicians to use in their offices. Ideally, this course would be shared to patients, in office, using a scalable QR code, as well as having the course link sent to the patient's phone number or email (or both), so that they have a copy of the link if anything happens where they can't find the course again on the web.

Physician Learning Program

Training Materials

For the purposes of this cholesterol information course, the two main groups we identified are instructors and those who would be trained, these groups are a) the family physician and b) the patients who will be using this cholesterol information course.

For Family Physicians

Before the course is given to patients the family physician would be required to take the course and complete the assessments, this would allow for physicians to understand the process of the course and assessments, and can be better equipped to answer questions from patients. By having the physicians take the course themselves, it also allows for them to identify possible problem areas that can be addressed and adapted if need be.

When a patient is identified by their physician as having high levels of low density lipoprotein (LDL) cholesterol through blood work, the physician would introduce and bring up the cholesterol information course, asking patients to go through the course and answer the questions honestly, adding that at the end there will be printable log sheets for them to track their progress. The doctor would request and encourage the patient to make a follow-up appointment to solely discuss the findings and learnings from the cholesterol information course and work with the patient to see what treatment plan (if any) is doable for them. This would include the patient's wishes, diet, exercise and potentially talking about medication options if necessary.

For the Patient Receiving the Cholesterol Information Course

After talking with their family doctor, the patient agrees that they will complete the course and will make a follow-up appointment in a week or two (or when is best for them) to discuss the results of the course and to discuss ways to manage their cholesterol safely.

The patient will open the link, and when they have the site open, there will be a welcome page, explaining what the course will be about.

The patient will be able to go through the course, at their own pace. Throughout the course, there will be little, complementary sections with questions that are associated with the modules being completed. The patient will need to check their knowledge after each module by doing a quick knowledge check to ensure they retain the information of the module

At the end of the cholesterol information course, the patient will need to do a quiz to show that they have understood the content, and must score 80% to pass the cholesterol information course. After the quiz, the patient will be able to print or send information from the course to themselves, as well as they will be provided with printable tracking logs.



Physician Learning Program

Prepare the Learners

For Physicians

From the perspective of the physician, they need to ensure that they can speak about the cholesterol information course and help to get patients, who might be hesitant, to show that the cholesterol information course is a benefit and will help them.

- It would be seen as beneficial if the physician does a test of the cholesterol information course tool to ensure accuracy and that navigation is user-friendly, and they can answer questions from patients
- Ensure that the physician has read the patient's lab work, and if abnormal, remembers to bring this up and to introduce the cholesterol information course in the appointment
- To remind patients to answer honestly and truthfully and to book a follow-up appoint in 1-2 weeks so that the physician and patient can discuss in greater detail about cholesterol and ways to manage it

For the Patient

For the patient, ways to ensure they have a good user-friendly experience using the cholesterol information course and feel empowered to take action regarding their cholesterol after taking the test include:

- Feeling supported and that the course is easy to navigate and use
- Having stable internet connection and having 20-30 minutes maximum to focus on the course, uninterrupted, or if this is not possible, time throughout their week to complete sections of the course
- Having a general health literacy vocabulary and can follow instructions, both from the physician verbally talking about the cholesterol information course and the prompts and questions embedded in the tool

Arrange Learning Space

From the Physician's Perspective

For the physician, they need to be aware that they have read the results of the lab tests and that the patient they have in clinic matches with the results of the lab.

Ensure that there is adequate time, at least 5-8 minutes, to talk about the cholesterol information course and the reasons they would want the patient to do it and answer any initial questions the patient might have. The physician should indicate that the cholesterol information course they are prescribing the patient is just a step in the right direction, ensuring that the patient and physician are on the same page as they journey through cholesterol health. The physician will need to ensure that the patient leaves the appointment

knowing what they need to do. The physician should also tell the patient that the course will take a total of 20-30 minutes to finish. Additionally, the physician needs to inform and encourage the patient to book a follow-up appointment to discuss the results of the cholesterol information course at the patient's earliest convenience - so they can take control of the patient's cholesterol. Ideally, the patient would book a follow-up before they leave the physician's office to ensure that they come back.

For Patients

For patients who will be filling out the cholesterol information course, they should be aware that it will take 20-30 minutes to finish and to answer the questions as honestly as possible. Additionally, it is recommended that they find a quiet space to finish the course, so they are able to fully understand the information being given.

Evaluation Plan

Formative Assessment

Patients

Before and after the patient finishes the learning course, they will be prompted to share their confidence about the topic. The patient will be informed that any and all answers will remain anonymous to encourage honest answers. The formative assessment will contain a question at the start of the course, and at the end, to help assess the confidence of each patient in regard to their initial knowledge on cholesterol and then their confidence in their knowledge after they have completed the course. This will help to assess the efficacy of the course.

Question that will be asked are:

- What is your confidence level from 1-5 on your knowledge of cholesterol after taking this course?

Physicians

Physicians will be asked to keep track of the number of the courses they give out to patients to complete. These numbers will be complied at the end of each month to assess whether the courses are being given out to a sufficient number of patients each month.

- How many patients have you given the course to?

Summative Assessment

Patients

After completing the course, at the end, the patients will be asked to complete a short survey asking them about the course and its ability to deliver information efficiently and effectively to patients. They will be prompted to answer the survey anonymously through a platform that PLP chooses to collect this information and responses.



Physician Learning Program

Questions That Will Be Asked

Reaction:

- The questions were given in a way that made them easy to understand
  - » Not at all
  - » Somewhat
  - » Yes
- The instructions given were clear and easy to follow
  - » Not at all
  - » Somewhat
  - » Yes
- The doctor prompted me to book a follow-up appointment after the course was done
  - » No
  - » Yes
- The images/ infographics were easy to understand and helpful in my learning about cholesterol
  - » Not at all
  - » Somewhat
  - » Yes
- What went well, and what could be improved in the course? Please be specific (Open ended question)
- Did you book a follow-up appointment with your physician?
  - » No
  - » Yes

Learning:

- I am more encouraged to try and improve my cholesterol health after I completed the course
  - » Not at all
  - » Somewhat
  - » Yes
  - » Extremely



Physician Learning Program

- The information given from the course has helped me understand the importance of a healthy lifestyle, what statins are, and the importance of cholesterol health and knowledge
    - » Not at all
    - » Somewhat
    - » Yes

Physicians

Physicians should keep track of how many patients book follow-up appointments after completing the course, in order to have a clear idea of how many patients are willing to learn more and make changes in their life after learning more about the importance of cholesterol, as well as the completion rate of the course. The physician should also be continually encouraging check ins with patients once a follow-up appointment has been made, after the completion of the course, in order to facilitate an ongoing relationship with the patient in regard to their cholesterol health. These questions should be delivered to the physicians every 6 months to a year in order to have a clear understanding of the efficacy of the course from a physician's point of view.

Questions to ask the physicians are:

- How many follow-ups have been booked as a result of the online course? (add only a numeric value)
- Have you encouraged more appointments past the initial follow-up to assess your patients and their ongoing plan for their cholesterol health?
    - » Yes
    - » No
- Are there any concerns or comments you have in regard to the course? (open-ended)

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